Conceptual Paper of the Trading Strategy: Dogs of the Dow Theory (DoD)

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Abstract

Investment objective of yielding higher return at lower risk is one of the challenges faced by participants (mainly investors) in share market. With an aim of overcoming this common challenge, past scholars have tested various trading strategies and even proposed new strategies but the outcomes are still puzzling. These high evolutions pertaining to trading strategies that occurred in investment world are covering several aspects such as fundamental features (ratios of the companies) and the economic variables yet the results were discourage due to mix results reported. Ample of causes could attribute to these situations where one of the most identifiable reason was unpredictability in global economic condition. Thus, this paper attempts to focus on the high yielding strategy of Dogs of the Dow Theory as one of the trading strategy in constructing portfolio in which this strategy are distinctive from the common high yielding approaches. This paper consists of several parts, namely the evolution of the trading strategies and the empirical evidences supporting Dogs of the Dow Theory as a trading strategy.

Keywords: trading strategies, portfolio selection, high-yielding strategy, dogs of the dow theory

JEL Classification: D92, G00, G10, G11, G14

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1 Introduction

Effective trading strategies defined as decisions made by investors to manage their investment in order to meet the goals and objectives in investing activities (Brown and Reilly, 2009). Trading strategies will benefits the investors from several aspects especially in gaining higher returns at lower risks. Further describes as the actions taken by companies and investors in generating desirable returns through designated portfolios, powerful and reliable trading strategies are also crucial and are required because of the competitions among companies in attracting potential investors (Ekaputra and Sukarno, 2012).

In investment world, returns are known as rewards for investors in bearing risks, whereas risks are the chances that the actual returns vary from the expected returns targeted by the investors (Basu and Chawla, 2010). Usually, risks fall under two categories, namely systematic risks (uncontrollable and non-diversifiable risks) and unsystematic risks (controllable and diversifiable risks). However, in relation to investment activities, only systematic risks (known as market risks) are considered due to the ability of portfolio to eradicate the unsystematic risks (known as unique risks) through diversification strategy (Kazi, 2008). William Sharpe (1964) and John Lintner (1965) first declared these ideas through the theory of Capital Asset Pricing Model (CAPM) and it had been widely accepted and practiced among scholars.

Diversification strategy in fact is one of the most effective techniques in reducing risks of investment since it prohibits investors from putting all eggs in one basket. The rationale behind this technique is the content of the portfolio whereby shares from various industries will be grouped together in constructing a portfolio. As a result, individual risk possess by each industry could be controlled due to varieties of shares from different industries that could provide higher returns at lower risks (Hoh et al., 2011).

2 Evolution of Trading Strategy

Numerous well-known and highly practiced trading strategies or theories have evolved where one of the fundamental theory in constructing a portfolio was known as Modern Portfolio Theory (MPT). Proposed by Harry Markowitz in 1952, the theory has since been expanded by William Sharpe (1964) and John Lintner (1965), which led to the development of Capital Asset Pricing Model (CAPM) before it was further enhanced solely by Stephen Ross in 1976 as the Arbitrage Pricing Theory (APT). The ideas behind these three theories vary yet it is still related to the construction of portfolios' strategy as a whole. The popularity of these theories also is conclusive since it still being practiced especially in academic field (Shamsabadi et al., 2012 and Zhang and Li, 2012) although their effectiveness are still being debated among past scholars due to mixed results reported.

Consequently, several new trading strategies had been developed in order to overcome the imperfections of these three theories and one of them is high-yielding strategy, which will be the focal point of this paper. One and the foremost well-known of high-yielding strategy is Dogs of the Dow Theory (DoD), which had been popularized by Michael O'Higgins and John Downes in 1991.

3 Literature Review

3.1 Modern Portfolio Theory (MPT)

Known as the father of modern portfolio theory, MPT was introduced by Harry Markowitz through the article of *"Portfolio Selection"* published in 1952. He suggested that investors are always expecting to be compensated whenever they are taking additional risk (Swisher and Kasten, 2005) whereby the issue of how to allocate funds among various assets was the ideology behind his

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book (Elton and Gruber, 1997). MPT also is associated with return and risk (variance) identification in determining the most efficient portfolios available.

Markowitz (1991) further stated that MPT is contrast from the theory of both firm and consumers in three major ways. Firstly, it focused on investors rather than the manufacturing firms or consumers and secondly, it is about economic agent who works under uncertainty. Lastly, MPT can be used to direct practice where investors who have a sufficient computer and database resources could benefits more from this theory (mainly, institutional investors).

Generally, MPT works under several assumptions where investors aim to increase the expected return throughout their investment and they make investment decision solely on the expected return and risk measurement. Besides that, they also are risk averse where they will accept greater risk only if they are compensated with higher return. Next, investors agreed to a single period investment horizon where there is no taxes and transaction cost in investment market. Finally, investors are able to access the information simultaneously (Ravipati, 2012 and Roychoudhury, 2007). Abundant of arguments occurred among past researchers on the reliability of MPT's assumptions especially those related to individual investors whom have unique investment behavior and risk tolerance (Wang and Zhang, 2012).

MPT is known as mean-variance analysis in choosing a portfolio of common shares. Mean-variance analysis also could be considered as a platform in creating and choosing of portfolio totally based on its expected performance of the investment and the risk tolerance of the investors (Fama and French, 2004 and Fabozzi et al., 2002). Pertaining to this analysis, past researchers again argued that investors tend to possess bias attitude in selecting shares under that portfolio since the high availability of other assets in market but yet, it is quite impossible to analyze them (Ravipati, 2012 and Hens and Hoppe, 2001). Thus, instead of share portfolio selection, Ravipati (2012) assumed that this analysis was more beneficial for asset allocation.

Markowitz had demonstrated two main beliefs under this mean-variance analysis where constant variance should be associated with maximum expected return while minimum variance must be linked with the constant expected return (Fama and French, 2004; Elton and Gruber, 1997 and Markowitz, 1991). Due to these two principles, the efficient frontier had been created whereby investors might choose their desired portfolios totally based on their risk preferences (Elton and Gruber, 1997 and Markowitz, 1991) since well-diversified portfolios are represented by the efficient frontier (Ravipati, 2012).

In general, MPT caused huge debates involving several issues among past researchers where they claimed that one of the major drawback of MPT is zero suggestion regarding the best portfolio out of the efficient portfolios available. As highlighted by past researchers, main issues that might arise is when investors might hold different efficient portfolios with different level of risk since MPT does not suggested directly the proper efficient portfolios for investors (Marling and Emanualsson, 2012; Rubinstein, 2002 and Roll, 1977). Besides that, although Curtis (2004) mentioned that markets are not concerned on the perception of investors towards risks, Swisher and Kasten (2005) believed that implementation of variance or standard deviation as a risk measurement in creating a portfolio is not proper because, investors tend to view risks in a different way.

Essentially, Markowitz realized that evolution or enhancement in his origin theory is needed when he found the difficulties in implementation of mean-variance analysis due to the tough calculation of mean-variance efficient portfolio especially when it involved a huge number of securities. Thus, together with William Sharpe, he proposed one solution to this problem, which was "single index model' (referred as one-factor model) and this idea had been incorporated into Capital Asset Pricing Model (CAPM) (Sullivan, 2006).

3.2 Capital Asset Pricing Model (CAPM)

Developed by William Sharpe (1964) and John Lintner (1965), CAPM was inspired by two papers, which were *'Portfolio Selection'* by Harry Markowitz in 1952 as well as *'The Cost of Capital, Corporate Finance and the Theory of Investment*' by Franco Modigliani and Merton Miller

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in 1958 (Sullivan, 2006 and Fama and French, 2004). Actually, the main focal of CAPM is to identify whether returns are statistically related to risks (beta) (Raza et al., 2011) and CAPM predicts that the positive function of beta, risk-free rate and expected market return is the expected return of assets (Verma, 2011). In CAPM, beta is used to identify the level of sensitivity of change in return of securities whenever there is a change in market return (Dzaja and Aljinovic, 2013).

In addition to existing assumptions under MPT, CAPM had created other several assumptions where the first assumption of this theory was all investors are able to borrow or lend money at the same interest rate (risk-free rate) regardless of amount involved. Secondly, investors are rational as well as risk-averse and as for third assumption, they select shares based on its expected return and risk of the available investment opportunities. Then, the fourth assumption is about trading securities, where no transaction costs involved and investors operate for the same period of time in which analysis is based on single period investment is the fifth assumption made by the theory. Next, the sixth and seventh assumptions are dividend as well as capital gains are taxed at the same rates and all the assets are perfectly divisible and highly liquid in the market. Lastly, despite of having all and equal information, investors are able to access to all information concurrently (Wang, 2013; Wang and Zhang, 2012; Jecheche, 2011 and Raza et al., 2011).

Despite of being viewed as one of the most important and well-known model in finance field, there are past researchers who assumed that most of the assumptions under CAPM are illogical and the best example is about the unlimited amount of borrowing and lending activities among investors. This is illogical since usually a short-term government security is treated as a risk-free rate security where the uncertainty is laying at the real rate of return since the inflation might have impact on it (Wang and Zhang, 2012).

High validity of CAPM could be established through a set of important implications that had been created (Jecheche, 2011). The first important implication is that investors calculated their expected return by only considering systematic risk (beta) since CAPM assumed that by holding well-diversified portfolio, unsystematic risk (unique risk) could be fully eliminated (Glogger, 2008). However, Ansari (2000) and Jagannathan and Wang (1993) argued that fully eliminated unique risks is impossible in investment since beta among securities vary on both individual securities and across business cycle in which there are securities that much more volatile than others. Thus, they affirmed that the ability of fully eliminate unsystematic risk in CAPM should not be rewarded in the market. These arguments could be further proven where although there are past studies who found a significant relationship between return and beta, there are in fact abundant of past researchers who reported an insignificant relationship between return and beta due to several reasons. For instance, most of the Iranian investors choose to participate in property market due to its higher return (Baghdadabad et al., 2010) whereas study performed by Verma (2011) in 18 developed markets found that the existence of interconnectedness between those countries caused an insignificant relationship between return and beta.

The second important implication in CAPM is high level of systematic risk (beta) should yield a higher level of return (Jecheche, 2011). This is aligned with the rule of thumb of "high risk, high return" practiced in investment because investors who are exposed to high degree of risk in investing activities will seek for a higher return. On the contrary to this implication, few past researches conducted found that higher risk (beta) do not lead to a higher return (Dzaja and Aljinovic, 2013; Yasmeen et al., 2012; Baghdadabad et al., 2010; Basu and Chawla, 2010; and Choudhary and Choudhary, 2010). As stated by Baghadadabad et al., (2010), many reasons could lead to this condition, among them are global and domestic economic situation, the government control and decisions plus the management teams (companies) at the decision making level also could influence the return earned by the investors.

The final important implication under CAPM is securities should be plotted on the Security Market Line (SML) whenever the risk (beta) is having a linear relationship with the return (Zhang and Li, 2012 and Jecheche, 2011). Linear relationship occurred when the risk and return are moving upward or downward simultaneously and there are diverse types of results reported by past studies. Firstly, besides a significant relationship between risk and return, the relationship between risk and

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return also is linear, which it is accordance to the assumptions of CAPM (Tang and Shum, 2003). Secondly, there are studies conducted that found a linear relationship between beta and return yet high beta is not associated with high return. Lastly, few past researchers found a nonlinear relationship between risk and return in their studies (Yasmeen et al., 2012 and Hodoshima et al., 2000) because of the fluctuation in return, as the economic conditions throughout the period study were unstable (Hasan et al., 2012; Choudhary and Choudhary, 2010 and Jarlee, 2007).

However, CAPM faced its major turning point regarding its credibility through the article critique wrote by Richard Roll in 1977 regarding the effectiveness of CAPM. Thus, it is believable that the drawback of this theory could be overcome through the introduction of the Arbitrage Pricing Theory (APT) (Jecheche, 2011; Cagnetti, 2002 and Shanken, 1982).

3.3 Arbitrage Pricing Theory (APT)

Initiated by Stephen Ross in 1976, APT was aimed as one of the best alternatives in overcoming the weaknesses of CAPM while still maintaining the basic messages under CAPM, but instead of a single factor model as possessed by CAPM, APT is more towards the multiple factors model (Baghdadabad and Glabadanidis, 2014). The core idea behind APT is similar items should not be sold at different prices and that the expected return of the securities could be modeled as a linear function of various macroeconomic factors (Basu and Chawla, 2012 and Jecheche, 2012). APT also is the first model that introduced the idea where the prices of assets are governed by several factors (Michailidis, 2009) and the importance of co-movement in asset returns was captured by numerous factors (Ferson and Korajczyk, 1995).

Several assumptions have been formed under APT. Firstly, the multifactor model will generate the return of the securities and secondly, the return generating process model is linear. Thirdly, APT requires a perfect competition in the market so that the opportunity for arbitrage does not exist and as the forth assumption, total number of assets should never exceed the total number of factors. The fifth assumption pointed out that there are no restrictions of short selling and finally, under obvious certainty, investors will favor for a more wealth compared to the less wealth (Gul and Khan, 2013; Basu and Chawla, 2012 and Jecheche, 2012).

Actually, APT encompasses many strengths as compared to CAPM where it requires less assumptions and it is suitable not only for investors and managers but also analysts and researchers especially in the academic line as it allows for more factors which can satisfies the conditions of empirical world. Finally, due to the multifactor model hold under APT, investors can also diversify their risk factors while implementing it (Jecheche, 2012 and Zhang and Li, 2012). This is because, securities react differently depending on the types of risks involved (Basu and Chawla, 2012), and due to its multifactor model, the explanatory power of APT is therefore should be better than CAPM (Cagnetti, 2002).

All of the above strengths have been proven through number of past studies, which found that APT performed better than CAPM in their researches. For example, study performed by Nguyen (2010) found that before the Asian financial crisis, APT factor of exchange rates and industrial growth rates were able to explain the return in emerging market of Thailand while CAPM single market model (beta) does not hold in that particular market. As for Febrian and Herwany (2007) who conducted study in emerging market of Indonesia for three different period (before, during and after Asian financial crisis), they found that APT was able to explain the return in all three different phases of economic condition being studied whereas CAPM was only suitable for economic downturn. Lastly, Cagnetti (2002) who conducted study in developed market of Italy also found similar results whereby APT with multi-factors was able to explain the return significantly compared to CAPM.

Nevertheless, APT has also several drawbacks that had been debated among past researchers since there are past researchers who found that CAPM performed better than APT. For examples, Zhang and Li, 2012; Widianita, 2009 and Paavola, 2006 reported that CAPM performed better than APT in China, Indonesia and Russia respectively, whereas study conducted by Gul and Khan (2013)

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and Faruque (2011) in Pakistan and Bangladesh verified that APT in not applicable at all. All of these situations was due to the APT's fundamental problem of not having a definite type of risk factor (Gul and Khan, 2013; Zhang and Li, 2012 and Paavola, 2006). In fact, it is a model that requires the users to identify the best risk factors that will affect the return without any restrictions.

Having variations of past findings concerning APT is very confusing. Thus, abundant of new theories had been initiated in order to improve the determination of the expected return where one of the most outdated yet reliable theory is Dogs of the Dow Theory (DoD).

3.4 Dogs of the Dow Theory (DoD)

Popularized by Michael O'Higgins and John Downes through their book of '*Beating the Dow'* in 1991, the Dow 10 Theory (commonly known as the Dogs of the Dow Theory (DoD)) is considered as one of the most well-known investment strategy in United States (US) with an aim of outperforming the market's performance. In constructing a portfolio, 10 shares with the highest dividend yield (DY) is selected from the Dow Jones Industrial Average (DJIA) whereby this portfolio need to be rebalanced and updated in a yearly basis. Prior to O'Higgins and Downes's book, the investment strategy of using high DY shares was first initiated by analyst John Slatter in 1988 through his article of '*Study of Industrial Averages Finds Stocks With High Dividend Are Big Winners'* where he found that the 10 highest-yielding shares of DJIA were able to outperform the DJIA statistically from 1973 to 1988. However, the DoD theory was only being acknowledged by investors through book written by O'Hoggins and Downes (Qiu et al., 2013 and Tissayakorn et al., 2013).

The *Dogs'* term under DoD refers to losers since those 10 highest yielding shares implies the lower prices currently possess by those shares (Tissayakorn et al., 2013). In reality, one of the criteria of DY strategy is earnings can go down while dividends can go up since investors tend to focus more on earnings. It indicates that the demand for respective shares will fall once there is a decrement in earnings reported by those companies, which will further reduce the shares prices (O'Higgins and Downes, 2000) since earnings' level is one of the main driver of the dividend's level (Andersson et al., 2010). Moreover, dividends also represent the real value of the company and thus, DoD theory actually is a strategy of constructing a portfolio comprises of undervalued or out-of-favored shares that deem to accelerate when the market is in a favorable condition (Ekaputra and Sukarno, 2012).

In details, unlike typical high-yielding strategies, O'Higgins and Downes (2000) claimed that DoD theory consists of several unique criteria. Firstly, the selection of shares in the DoD portfolio was based on the blue chip shares in the market, which represent the main sector and economic condition of that particular country. This is important because it represents the overall picture of market performance and any concentrated risks possessed by individual industry could be eliminated (Rowlett, 2012). Thus, shares involved usually are the biggest and strongest companies, where it represents the most liquid and well-known companies that have a sustainable growth, vast resources and excellent reputation (Jeong et al., 2008). Nevertheless, Damodaran (2004) declared that investing in both matured and young companies will benefits investors in different perspectives and the best reason is companies with less followers (young companies) tend to be misvalued and sometimes there is a lack or no involvement from institutional investors at the early stage. Under this condition, the young companies will be able to attract more potential investors (individual and institutional) once they become larger and this will benefits the existing shareholders. The most interesting part is that, the existing shareholders could demand for a better payoff throughout the growth processes of those respective companies. Study conducted by Banz (1981) in US also proved that limited diversification caused by lack of information regarding the small companies could provide a significant abnormal return towards the shareholders.

Secondly, DoD is classified as a short-term investment strategy due to the requirement of yearly rebalanced and updated the DoD portfolios. As stated by O'Higgins and Downes (2000), the main problem of long-term strategy (buy and hold strategy) is liquidity where once investors are in

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need of cash, they might oblige to sell the best shares (best performers) that they possess or else they have to bear losses by selling the underperforming shares. Then, investors also might need to forego the chances of substituting the current best shares if they choose to involve in long-term horizon. However, Damodaran (2004) affirmed that compared to short-term period, shares are less risky in the long-term horizon because the bad year (shares performed poorly) might be compensated by the good year (shares performed greatly).

Lastly, the original version of DoD theory involves ten shares in each portfolio where O'Higgins and Downes (2000) affirmed that DoD is a simple theory due to the same method applied in constructing the portfolio. It is actually suitable mostly for conservative investors since number of shares involved is considerably huge. Hence, institutional and individual investors could implement this theory due to its simplicity and applicability compared to other strategies (Ekaputra and Sukarno, 2012). Contrary to this statement, Damodaran (2004) argued that the return of a portfolio comprising of only ten shares could be misleading since there are thousands of shares available in share market and usually, the Dow dogs are much riskier than the other shares where the higher return gained was due to that higher risk. He also asserted that the criteria of high dividend shares is suitable for risk averse investors but the classification of loser shares under DoD are appropriate for risk seeker investors. The reason is due to the history of the companies where they are promising interesting opportunities to the investors who are willing to take risk only after the company experienced bad performance in the past.

DoD theory is categorized as a contrarian strategy since in general, one of the main contrarian indicator is high DY. It shows that the investors doubt about the future earnings of that company because high yield is associated with lower share prices (O'Higgins and Downes, 2000). Companies are at the bottom of the business cycle if their shares encompass a lower market price due to high DY and those prices most likely will increase faster than low yield shares, which will further benefits the investors (Ekaputra and Sukarno, 2012). This is similar with the study conducted by Andersson et al., (2010) who mentioned that besides a lower beta, high yielding shares usually have a higher alpha, which means that those shares tend to provide higher return than what have been suggested by their level of risk. On the negative side of this contrarian strategy, Damodaran (2004) claimed that the contrarians investors tend to rely more on their instincts in making judgment pertaining to share market movement where there are two simple arguments used by this type of investors. First of all is '*it is always darkest before dawn',* which means that the best time to buy the shares is after bad news have pushed down the price because through bargaining process, the possibility for its market price to rally is higher. Second arguments is '*lower-priced shares are cheaper',* where shares that have drop a lot in prices often being traded at a lower prices and there are investors who felt that lower-priced shares are cheaper and attractive due to bargain opportunity, regardless of the threat that the prices might fell drastically on the same period.

Apart from the contrarian categorization, DoD theory also is known as a value investment strategy due to its high-yielding shares. According to Visscher and Filbeck (2003), high DY and low in price-to-book ratios, price-to-earnings ratios as well as expected growth rate are the characteristics of value shares whereas growth shares are low in DY but high in price-to-book ratios, price-to-earnings ratios as well as expected growth rate. The logic behind this condition is that investors usually will overreact negatively towards undesirable company financial news (result in downward market price movement) and positively towards superior company financial news (result in upward market price movement). Here, the former overreaction will create value shares while the latter overreaction will form growth shares.

There are many differences between value and growth investors but the major variation between those two is regarding the conformation with Efficient Market Hypothesis (EMH). As for value investors, they are against the EMH theory since they rely more on their personal judgment up to the level of when they assumes that there are mispriced in the shares. Usually, the indications of underpriced shares are a good buy signal while a good sell signal comes from overpriced shares. They considered this as a reward since they are participating in the depressed shares by substituting them from companies that having difficulties. This will further help to boost the share prices and

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sell them off once they are satisfied with the new price level. As for Hoh et al., (2011), they stated that this type of investors is categorizing as active investors who performing active strategy in constructing their portfolios. In contrast, growth investors are in the same flow of EMH where they assume that the current market prices are reasonable since it reflects all the currently knowable information about the company. Here, growth investors simply enjoy the rewards which already available for them. Thus, this group of investors is known as passive investors since they are applying passive strategy in constructing portfolios.

In general, investing and constructing portfolios consisting of high yielding shares are believable to benefit the investors from several aspects (Henne et al., 2009). Mainly, it is related to stable positive returns in uncertain markets (Henne et al., 2009) since one of the aims of high yielding strategy is to protect the investors from the downside risks by obtaining that huge return (Franken, 2012). This was supported by Alenius (2011) and Safari (2009) who found that DoD strategy was able to protect investors from the downside risks in Finland and Malaysia respectively since the DoD portfolio had outperformed the market in both bull and bear market. In addition, Barron (2001) found that during market crashed in Canada, high yield portfolio was able to act as inflation hedge since its value was rising and this portfolio was less volatile although the market was in a bad condition. Next, controlling overinvestment problems also could be one of the reason of why investors seek for high dividend income (Dong et al., 2005) as dividend are paid from real earnings in which questionable financial reporting practices could be avoided (Tripathi and Aggarwal, 2012). In fact, there are managers who are not competent in managing cash due to poor investment prospect and thus, high dividend declared could help to control this problem (Damodaran, 2004).

On behalf of the companies, declaring high dividend sometimes will be a good choice especially in a country such as Malaysia that impose corporate tax on retained pre-tax profit whereby declaring high dividend clearly will reduce the tax burden of the companies (Foong et al., 2007). Next is regarding the behavior of investors who viewed dividend as a signal of the companies' performances (financial stability). Essentially, DY is very informative (Montgomery, 2013) since reducing or cutting the dividend is a negative signal whereby it indicates that the companies are having extensive and long-term financial problems (Karpavicius, 2014 and Damodaran, 2004). Research conducted by Karpavicius (2014) also discovered that companies tend to maintain their dividend level in order to maximize their share prices even though in reality, the companies do not intend to use the dividend declared as a signal device for investors. This is because, the smoothness in paying dividend are positively related to the movement of share prices as well as the effectiveness of future performance of the companies. On top of that, higher share prices also will lead to higher confidence level among investors, which in turn will positively affect the spending of the investors and growth of the companies simultaneously (Montgomery, 2013). This situations happen in emerging market of Thailand where research conducted by Jiranyakul (2011) in Thailand proved that investors care more for dividend payment over capital gain since dividend return demonstrated the larger size and more highly significant in the risk and return relationship.

Investing in high-yielding shares also consists of several drawbacks and one of them is inability of valuing the companies that paying no dividend to their investors because cutting or declaring no dividend could happened in both big and small companies (Montgomery, 2013). As mentioned by Ghosh (2010), bigger companies in India always pay more dividends yet they also are the one that tend to cut dividend whenever they have problems. Then, high DY also does not inform the investors regarding the level of expected return from the capital gain and investors are responsible to estimate the intrinsic value of the shares. This is important since investors can identify whether the share traded is cheaper or expensive compared to its current price. Lastly, high DY does not indicates the economic return that investors could earn from owning a share since it is influence by several unpredictable factors (Montgomery, 2013).

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Commonly, applying DoD strategy could be done through two major approaches which further lead to two major effects, namely statistically effect (considering none of risks, taxes or transaction costs) and economically effect (considering either risks, taxes or transaction costs).

Historically, the first research conducted specifically for DoD was done by O'Higgins and Downes (1991) with respect to DJIA whereby the results reported that the DoD portfolio able to outperform the market statistically. As for academic purposes, McQueen, Shields and Thorley conducted the first research in 1997 (Qiu et al., 2013). Studying the DJIA for 30 years from 1946 to 1995, McQueen et al., (1997) first found that the Dow 10 outperformed the DJIA statistically. However, when they adjusted for risk and transaction costs as well as taxes, the Dow 10 then underperformed the DJIA, which further proven that Dow 10 were not able to beat the DJIA economically. The reasons are due to data mining and nonexistence of market anomaly.

Results found by McQueen et al., (1997) was supported by Hirschey (2000) who found that abnormal losses was recorded once the transaction cost and taxes was being considered and he claimed that this strategy was ineffective due to data error collected by O'Higgins and Downes (1991) throughout their study. Due to this complexity, Prather and Webb (2002) had decided to come out with a similar study conducted by O'Higgins and Downes (1991) in order to identify whether there was data error occurred as claimed by Hirschey (2000). As a result, they found that there was market anomaly that permitted an abnormal return generated and data error was not the factors that lead to the superior performance of this strategy.

Following those arguments among past scholars, number of researchers had come out with an attempt to test the applicability of DoD strategy in different share markets globally. Among the latest researches focusing solely on statistically effects were conducted by Rowlett (2012) as well as Ekaputra and Sukarno (2012), who found that DoD portfolios were outperforming the market statistically in Thailand and Indonesia respectively. Then, studies conducted in Hong Kong by Mingyue, Hong and Yu (2012), in Australia by Alles and Sheng (2008) and in US by Clavenger and Baker (2004) as well as Prather and Webb (2002) reported similar results where DoD portfolios outperformed the market statistically.

Nevertheless, there are researchers who implement this approach yet the outcomes are under the conditional basis, such as time horizon of holding the DoD portfolios. Actually, despite of study performed by O'Higgins and Downes in US, research executed in China by Wang et al., (2011) also found that the abnormal return could be generated in a shorter holding period. Conversely, contradict to the basis of DoD theory, Clemens (2013), Qiu et al., (2013), Tissayakorn et al., (2013), Alenius (2011), Kazi (2008) and Clevenger and Baker (2004) claimed that DoD portfolio could beat the market and generated higher abnormal return only in the long-term horizon.

Then, there were also studies focusing on the effectiveness of DoD strategies in different market condition (bull and bear market). Essentially, the DoD strategy should be able to protect investors from downside risk but contrary to this statement, Broberg and Lindh (2012) and Sahu (2001) reported that this strategy was effective only in rising market (bullish market). Finally, concerning this statistically significant, Domian et al., (1998), had conducted a study by comparing the performance of high-yield and low-yield portfolios based on pre-crash (1964-1986) and post-crash (1989-1997) period in US. They found that both portfolios reported mixed results since those portfolios recorded good and poor results individually throughout the period but the most highlighted point was that, high-yield portfolio was able to beat the market only for several months in both period before it underperformed the market completely in both period. Thus, the researchers affirmed that DoD strategy was not effective in US and the main reason stated was the absent of market anomaly since this strategy had been popular and widely implemented.

Next is about the implementation of more advance method (statistically and economically effect) where plenteous of researchers had duplicate this strategy that incorporated both developed and developing markets. Unlike results reported by McQueen et al., (1997), most of the recent researchers found that the DoD strategy was significant in both statistic and economic condition. The examples of most recent study was conducted in the developing market by Yan et al., (2015), Safari (2009), and Wolmarans (2004) who reported that DoD portfolios outperformed the market

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statistically and economically in Taiwan, Malaysia, and South Africa respectively. Parallel results also were reported in developed markets by Qiu, et al.,(2013), Andersson et al., (2010), Brzeszczynski et al., (2008), Lemmon and Nguyen (2008), Brzeszczynski and Gadjka (2008, 2007) and Visscher and Filbeck (2003). All the researchers revealed that abnormal return could be gained through DoD portfolio since it outperformed the market statistically and economically in Japan, Swedish, British, Hong Kong, Poland and Canada respectively.

In contrast to the above findings, number of researchers found the insignificant results between DoD strategy and abnormal return whenever risk, taxes, transaction costs or commissions being included. For instance, study conducted Turkish confirmed that DoD portfolio underperformed the market in both statistically and economically where they further found of no evidence that this DoD strategy was less risky than the buy and hold strategy (Prather et al., 2011). As for Dahlstedt and Engellau (2006) who performed study in Nordic Stock Market, they stated that the DoD strategy underperformed the market in both statistically and economically as in that particular market, high DY was explained by decrement in stock prices rather than increment in dividend paid. Next, studies conducted by Henne et al., (2009) in German Stock Market found insignificant relationship between DoD strategy and abnormal return. However, high-yield share is able to decrease the risk of investing but this influence fall substantially if the degree of diversification escalates. Finally, Leal et al., (2000) who conducted a research of DoD strategy with regard to developed markets of US versus developing market of Brazil stated that the results found were quite discouraging. This is because, DoD strategy was clearly ineffective to be implemented in Brazil but in US, the evidence of effectiveness was limited since DoD was significant statistically but not economically. It further suggested that DoD strategy can add value as investment strategy in US yet not suitable for small investors.

4 Methodology Used in Previous Research

The application of DoD strategy is simple yet attractive since it is based solely on the blue chip shares as listed on Top 30 Components or market index in any share markets around the world. The origin procedures of constructing, reviewing and updating portfolios are done on yearly basis whereby ten shares with the highest DY are chosen. Every year, shares with the highest DY will be included into the DoD portfolio and the analyzing process of assessing the return generated by the DoD portfolios will be done through comparison of the performance in terms of return between DoD portfolios and the market (O'Higgins and Downes, 2000).

Estimating the abnormal return earned could be performed in two ways, either including or excluding the transaction costs, taxes and risks. Although the inclusion is believable to give the best results and views regarding the DoD's effectiveness, abundant of past researchers decided to exclude those factors. The reasons could be risks factors (especially economic factors) that unpredictable in nature and regular changes in taxes policy implement by government, which might complicate the process of gauging the abnormal return.

5 Conclusions

Investors are interested in making money from their investments. Their aim is to have high return and low risk. Many trading strategies have been developed in an attempt to maximize their investment returns and among the popular trading strategies available are CAPM and APT. Actually, DoD is one of the trading strategies that related directly with DY basis in which previous studies have reported mixed results pertaining to the effectiveness of DoD yet the highlighted point is that, results documented with respect to individual developed and developing markets are vary. Thus, claiming this DoD only effective either in developed or developing market is fictional due to those whole situations.

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This paper attempts to introduce and acknowledge the existence of this simple yet attractive trading strategy that is rarely being implemented among investors although it had been established for more than two decades.

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